

A classification of biologically unavoidable sequences

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Abstract

Alexander proved that every eventually periodic sequence occurs in every infinite gendered population satisfying finite roots, finite outdegree and finite birthdate sublevels, and asked whether any other sequence has this property. We construct an avoiding population for each non-eventually-periodic sequence over every finite alphabet. For a binary target, the population has two roots, exactly two parents per non-root and at most two children per vertex. A nonnegative decreasing path offset proves avoidance. The construction gives a classification by eventual periodicity. We also characterize universality within the constructed binary family and deduce finite-observation and computability obstructions. A Lean development checks the negative construction and its population axioms; the verification scope is stated explicitly. No literature priority is claimed.

1. Population axioms and the classification statement

Fix a nonempty finite alphabet A . Following Alexander [1, Definition 1], a population consists of a directed graph (V, E) , edge labels in A , and birthdates $t : V \rightarrow \mathbb{R}$ such that:

1. There are finitely many vertices with no incoming edge, called roots.
2. Every vertex has finitely many children.
3. For each real R , the set $\{v : t(v) \leq R\}$ is finite; every edge $u \rightarrow v$ satisfies $t(u) < t(v)$.
4. The vertex set is infinite.
5. Every non-root has an incoming edge of each label in A .

Each ordered pair of vertices carries at most one edge and one label. A sequence $s : \mathbb{N} \rightarrow A$ is realized if there exists a path $v_0, v_1, \dots$ with k th edge labelled $s(k)$. There is no requirement that v_0 be a root. A sequence is biologically unavoidable when every population over its alphabet realizes it. It is eventually periodic when there exist integers $K \geq 0$, $p > 0$ such that $s(k+p) = s(k)$ for all $k \geq K$. Throughout this paper, aperiodic means not eventually periodic.

Theorem 1 (classification). A finite-alphabet sequence is biologically unavoidable if and only if it is eventually periodic.

The positive direction is Alexander’s Theorem 10 [1], proved again in Section 4 with an explicit boundary repair. The negative direction is the construction in Sections 2–3. It answers the final question of [1, Section 6]. The other questions there about universal populations and ordinal characterizations are not all claimed to be resolved by this theorem.

2. An explicit binary avoiding population

Fix $s : \mathbb{N} \rightarrow \{0, 1\}$ and put

$$r(2j) = s(j), \quad r(2j+1) = 1 - s(j).$$

Let P_s have vertices $\mathbb{N}$ and birthdate $t(v) = v$. For every $v \geq 2$ put the following two incoming edges, and no others:

$$\begin{aligned} v-1 &\longrightarrow v \quad \text{with label } r(v), \\ v-2 &\longrightarrow v \quad \text{with label } 1-r(v). \end{aligned}$$

There is no edge $0 \rightarrow 1$. The roots are exactly 0 and 1. Vertex 0 has one child, and each vertex $u \geq 1$ has two children. Every non-root has two distinct parents, with complementary labels. Birthdates increase strictly, their sublevel sets are finite, and the graph is infinite. Thus these are populations in the precise edge-labelled sense of [1].

Claim. If P_s realizes s , then s is eventually periodic.

Proof. Suppose $v_0, v_1, \dots$ is an infinite matching path. Each increment is 1 or 2. Inductively,

$$v_k \geq 2k.$$

Indeed, if $v_k \geq 2k+1$, the next vertex is at least $2k+2$. If $v_k = 2k$, an increment of 1 would have label $r(2k+1) = 1 - s(k)$, so is impossible. At $k=0$ this edge is absent; the same conclusion holds. The base case is $v_0 \geq 0$.

Consequently $d_k = v_k - 2k$ is a nonnegative integer sequence with increments in $\{-1, 0\}$. It eventually stabilizes at some $d \geq 0$. From some K onward every path increment is therefore 2, so

$$s(k) = 1 - r(2k + d + 2), \quad k \geq K.$$

If $d = 2e + 1$, this says $s(k) = s(k + e + 1)$, giving positive eventual period $e + 1$. If $d = 2e$, it says $s(k) = 1 - s(k + e + 1)$; applying the relation twice gives period $2e + 2$. Either case forces eventual periodicity. $\square$

Thus **every non-eventually-periodic binary target is avoided by its own P_s , from all starting vertices**. The graph depends on the target; this does not produce one graph avoiding every aperiodic sequence.

3. Finite alphabets and vertex genders

Let A be finite and $s : \mathbb{N} \rightarrow A$ not eventually periodic. At least one indicator projection $\pi : A \rightarrow \{0, 1\}$ has a non-eventually-periodic image $\pi \circ s$. Otherwise all finitely many symbol indicators would be eventually periodic. Beyond the maximum of their starting thresholds, a common multiple of their positive periods would be a period of s .

The choice of an aperiodic indicator here is existential. No algorithm for selecting it from an arbitrary sequence program is asserted. The binary construction itself is uniform relative to its target; the computability reductions in Section 5 use only that binary construction.

Construct the binary population $P_{\pi \circ s}$. Replace each vertex v by the copies (v, a) , $a \in A$. For every binary edge $u \rightarrow v$ with label c , add

$$(u, a) \longrightarrow (v, b) \quad \text{with label } a \quad \text{for all } a, b \in A \text{ satisfying } \pi(a) = c.$$

Give each copy birthdate v . There are $2|A|$ roots, at most $2|A|$ children per vertex, and exactly one incoming parent of each label at every non-root. The other axioms are inherited. An s -path would project to a $\pi \circ s$ -path in the binary population, which is impossible.

Moreover, label a is the source copy's second coordinate: assigning gender a to (v, a) makes this a **vertex-gendered** population. For a binary alphabet this yields four roots and at most four children per vertex. The smaller two-root/two-child construction uses the more general edge-labelled definition.

Thue–Morse example. Put $t(k)$ equal to the parity of the number of ones in the binary expansion of k . Then $t(2k) = t(k)$ and $t(2k + 1) = 1 - t(k)$, so the auxiliary sequence r equals t itself. The incoming labels at v are simply $t(v)$ and $1 - t(v)$.

For completeness, t is not eventually periodic. If $p > 0$ were an eventual period, then for every sufficiently large m , $t(2^m - p) = t(2^m) = 1$. But $2^m - p$ is the m -bit complement of $p - 1$, so

$$t(2^m - p) = (m \bmod 2) \text{ xor } t(p - 1),$$

which alternates with m . This is a contradiction.

4. The positive direction and its boundary repair

This section recalls the mathematical input from Alexander [1, Proposition 9 and Theorem 10]. It is included to make the classification argument self-contained and to clarify a deletion step.

Let V_j consist of vertices reachable in at most j edges from a root. Each V_j is finite by finite roots and finite outdegree. Every vertex has finitely many ancestors, by the birthdate axiom, and every vertex descends from a root: a backward path must terminate inside its finite ancestor set.

Proposition 2 (Alexander). Every periodic sequence is unavoidable.

Proof. Let s have period $p > 0$ and let $w = s(0) \cdots s(p - 1)$. Starting at any vertex v , repeatedly try to select a length- p path backwards whose forward labels are w . Whenever a full block can be selected, retain it and repeat from its first vertex. This process must terminate because all selected vertices are distinct ancestors of v . If selection stops at u , some partial backward choice of fewer than p parents reaches a root, so $u \in V_{p-1}$. Otherwise all p choices would be available by the incoming-label condition. Thus there is a path from V_{p-1} to v labelled w^m for some $m \geq 0$.

The lengths of these paths are unbounded as v varies: a finite set with finite branching reaches only finitely many vertices within any bounded number of steps, whereas the population is infinite. The forest of all s -matching finite paths starting in the finite set V_{p-1} is finitely branching and has

unbounded height. König's lemma gives an infinite s -matching path. This proves the proposition. $\square$

Proposition 3 (finite prefixes). If s is unavoidable and w is finite, then ws is unavoidable. Conversely, unavoidability of ws implies that of s .

Proof. The converse follows by deleting the first $|w|$ edges of a realizing path. For the forward implication, take any population P and write $k = |w|$. There is nothing to prove when $k = 0$. Delete V_{k-1} , then delete every remaining edge entering the finite boundary $V_k \setminus V_{k-1}$. Those boundary vertices become roots. Every vertex outside V_k retains all its original parents, since a parent in V_{k-1} would put its child in V_k . All population axioms therefore hold in the modified graph P' , which is infinite because only finitely many vertices were removed.

Realize s in P' . Its starting vertex lies outside V_{k-1} in P . Choose k parents backwards in P with labels spelling w forwards. No root can be encountered before completing those k choices, since that would put the starting vertex in V_{k-1} . Prepending this path realizes ws in P . $\square$

Applying Proposition 3 to a periodic tail proves the positive direction of Theorem 1. Merely taking the induced graph after deleting V_{k-1} , as in the published proof, need not preserve incoming-label coverage. For example, put two roots a_0, b_0 at time zero and pairs a_i, b_i at each later integer time, each normally receiving a label-0 edge from a_{i-1} and a label-1 edge from b_{i-1} . Replace the label-0 parent of a_2 by a_0 . After the roots are deleted, a_2 remains a non-root with no label-0 parent. The additional boundary-edge deletion above repairs that step without changing the theorem.

5. Universality and computability in the witness family

The construction has a second exact property:

$$P_s \text{ realizes every binary sequence} \iff s \text{ is eventually periodic.}$$

The negative direction is the binary avoidance theorem in Section 2. If s has eventual period p , then r has eventual period $q = 2p$. Form a finite residue multigraph modulo q , with the two incoming transitions into residue a , labelled by the eventual $r(a)$ and its complement, and marked with delays 1 and 2. Every state has an incoming transition of each label. Every finite word can therefore be read backwards; finite branching and finitely many possible starts give an infinite path for every infinite word. Starting sufficiently late in P_s , lift each transition by its marked delay. This realizes the word in the actual population. The marked delays retain the information needed to lift the quotient path.

The graph through vertex N depends only on $s(0), \dots, s(\lfloor N/2 \rfloor)$. Any such prefix can be continued with zeros or with Thue–Morse. The resulting populations agree on that entire finite region, but one is universal and the other is not. This also obstructs any finite adaptive sequence of local edge queries.

Even a program describing the graph does not decide universality. Given a machine M , let $s_M(k)$ be the k th Thue–Morse bit if M has not halted within k steps, and zero otherwise. This sequence and its graph are total computable objects. The graph is universal exactly when M halts. Conversely, using zeros until a halt is detected and then starting a Thue–Morse tail gives a universal graph exactly when M never halts. Thus neither universality nor non-universality is semidecidable on these presentations, even with the promise that all population axioms hold.

For a fixed aperiodic s , breadth-first matching from any vertex terminates with an empty set by Section 2 and König’s lemma. There is no bound uniform in the target even for starting vertex 1: a target with $N + 1$ initial zeros and an aperiodic tail admits N matching edges along $v_k = 1 + 2k$. Obtaining useful explicit bounds for the fixed Thue–Morse target remains a quantitative question.

6. Verification and reproducibility

The Lean project defines the actual edge relation and eventual periodicity, proves the binary path implication, proves the finite-alphabet indicator reduction, and constructs the lifted population. The final endpoint is `BiologicalAvoidance.finite_alphabet_real_avoidance`, using explicitly stated real-birthdate population axioms. Consult `verification/lean.json` and `verification/lean.log` for the check result for this release and exact source hashes.

The formal statement quantifies over every finite alphabet and every target that is not eventually periodic. It constructs a graph on the natural numbers times that alphabet, with real birthdates, satisfying all displayed population conditions and containing no infinite matching path from any starting vertex. The label predicate is functional on ordered vertex pairs. The formal proof does not assume an aperiodic binary projection or a population-existence lemma; those are proved internally.

Alexander’s positive direction, the proof that Thue–Morse is aperiodic, and the universality/computability corollaries in Section 5 remain prose proofs. The classification as a whole is not presented as an end-to-end Lean theorem. `STATEMENT-AUDIT.md` explains the correspondence and records that this audit was performed by the same AI system, not an independent reviewer.

The standard-library Python programs in `checks/` independently transcribe edges and matching paths and check finite periodic-template lifts. These are diagnostic controls, not proofs of infinite claims. For example, failure to find a barrier within a fixed finite horizon is recorded as unresolved by that computation, never as an infinite path.

7. Attribution, provenance and scope

Samuel A. Alexander’s positive theorem and the original population model are prior work [1]. A human collaborator selected the paper and directed the research. OpenAI Codex (GPT-6 family; no more specific deployment identifier is asserted) developed the new construction, arguments, formalization, exact checks and exposition.

Searches described in `literature-review.md` did not verify an earlier classification, but do not establish priority. The short computational reductions are consequences of the construction and the standard halting obstruction; their inclusion is not a claim that these methods are novel. No physical or biological prediction is inferred beyond the population model’s stated axioms.

References

- [1] Samuel A. Alexander, *Biologically unavoidable sequences*, Electronic Journal of Combinatorics **20**(1) (2013), P31. DOI: 10.37236/3035. Published PDF. arXiv:1212.0186. Theorem numbers above refer to the published version.